

Exercise #1

A 3R planar robot is subject to hard joint velocity limits $|\dot{q}_i| \leq V_i$, for $i = 1, 2, 3$, with $V_1 = 1$, $V_2 = 1.5$, and $V_3 = 2$ [rad/s]. At the current configuration, its task Jacobian is given by

$$J = \begin{pmatrix} -1 & -1 & -0.5 \\ -0.366 & -0.866 & -0.866 \end{pmatrix}$$

and the task requires a Cartesian velocity $v = (2 \ 1)^T$ [m/s]. Apply the SNS algorithm to find a feasible solution $\dot{q} \in \mathbb{R}^3$ with the least possible norm, including task scaling if needed.

The solution of the SNS method is given by

$$\dot{q}_{SNS} = (JW)^{\#} s v + (I - (JW)^{\#} J) \dot{q}_N$$

$$A = J \quad C = (JW)^{\#}$$

$$B = JW$$

A solution through the classical $J^{\#}$

$$\text{provide } \dot{q} = J^{\#} v = \begin{cases} -1.335 \\ -0.738 \\ 0.148 \end{cases}$$

we saturate $\dot{q}_1$

$$W = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \quad \dot{q}_N = \begin{pmatrix} 1 \\ 1.5 \\ 2 \end{pmatrix}$$

$$\dot{q}_{SNS} = \begin{pmatrix} 0 & 0 \\ -2 & 1.154 \\ 2 & -2.3 \end{pmatrix} s \begin{pmatrix} 2 \\ 1 \end{pmatrix} + \begin{pmatrix} 1 & 0 & 0 \\ -1.573 & 0 & 0 \\ 1.154 & 0 & 0 \end{pmatrix} \begin{pmatrix} 1 \\ 1.5 \\ 2 \end{pmatrix} = s \begin{pmatrix} 0 \\ -2.846 \\ 1.7 \end{pmatrix} + \begin{pmatrix} -1 \\ 1.573 \\ -1.154 \end{pmatrix} = \begin{pmatrix} -1 \\ -1.273 \\ 0.546 \end{pmatrix}$$

Exercise #2

Determine all the conditions on the constant parameters a , b , c and d , under which the following (linear) equations can be considered the dynamic model of an actual 2-dof robot:

$$M\ddot{q} + g = u, \quad \text{with } q \in \mathbb{R}^2, u \in \mathbb{R}^2,$$

where

$$M = \begin{pmatrix} a & b \\ b & c \end{pmatrix}, \quad g = \begin{pmatrix} 0 \\ d \end{pmatrix}.$$

Any guess about which type of robot this could be? Suppose now that only one generalized coordinate is actuated by a command $u_a \in \mathbb{R}$, while the other is passive. Which component of u should be actuated in order to guarantee the existence of an equilibrium? If we choose a value $u_a > 0$, what would be the instantaneous acceleration of the other (passive) coordinate?

On the first link, the gravity does not perform work, since $g_1 = \frac{\partial U}{\partial q_1} = 0$, we know that U does not contain q_1 , so if joint 1 is revolute, it's z axis is vertical, if it's prismatic, is on the horizontal plane.



Since $g_2 = d$, U depends linearly from q_2 : $U = dq_2$ and probably $d = m_2 g_0$ so the second joint is prismatic and vertical.

We have two possibilities

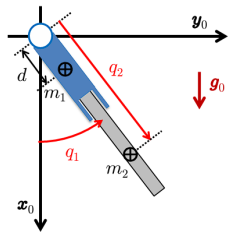


In both cases, to have an equilibrium we need to actuate joint 2.

$$\begin{cases} 2\ddot{q}_1 = 0 \\ c\ddot{q}_2 + d = u_a \end{cases} \quad \text{if } u_a > 0 \quad \begin{cases} \ddot{q}_1 = 0 \\ \ddot{q}_2 = (u_a - d) \frac{1}{c} \end{cases}$$

Exercise #3

Consider the RP robot in Fig. 1, moving in a vertical plane. The prismatic joint has a limited range $d \leq q_2 \leq L$. Derive the gravity term $g(q)$ in the dynamic model and find the expression (in symbolic form) of a constant $\alpha > 0$ that bounds $\|\partial g(q)/\partial q\|$ for all q within the robot workspace.



$$P_{c1} = \begin{cases} dc_1 \\ ds_1 \end{cases} \Rightarrow U_1 = -m_1 P_{c1}^T \hat{g}_0 = -m_1 g_0 d \cos(q_1)$$

$$\hat{g}_0 = (\underbrace{g_0}_{\hat{g}_0} \ 0)^T \quad P_{c2} = \begin{cases} q_2 c_1 \\ q_2 s_1 \end{cases} \Rightarrow U_2 = -m_2 g_0 q_2 \cos(q_1) \Rightarrow U = -g_0 ((m_1 d + m_2 q_2) \cos(q_1))$$

$$\Rightarrow \frac{\partial U}{\partial q} = \begin{cases} g_0 (m_1 d + m_2 q_2) \sin(q_1) \\ -g_0 m_2 \cos(q_1) \end{cases} \Rightarrow$$

$$G = \frac{\partial g}{\partial q} = \begin{pmatrix} g_0 (m_1 d + m_2 q_2) c_1 & g_0 m_2 s_1 \\ g_0 m_2 s_1 & 0 \end{pmatrix} \quad \text{i need } G^T G$$

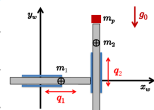
$$G^T G = \begin{bmatrix} [g_0 (m_1 d + m_2 q_2) c_1]^2 + [g_0 m_2 s_1]^2 & g_0 (m_1 d + m_2 q_2) c_1 g_0 m_2 s_1 \\ g_0 (m_1 d + m_2 q_2) c_1 g_0 m_2 s_1 & [g_0 m_2 s_1]^2 \end{bmatrix}$$

$$\det \begin{pmatrix} \lambda - [g_0 (m_1 d + m_2 q_2) c_1]^2 + [g_0 m_2 s_1]^2 & -g_0 (m_1 d + m_2 q_2) c_1 g_0 m_2 s_1 \\ -g_0 (m_1 d + m_2 q_2) c_1 g_0 m_2 s_1 & \lambda - [g_0 m_2 s_1]^2 \end{pmatrix}$$

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Exercise #4

The Cartesian robot in Fig. 2 has the two links respectively of mass m_1 and m_2 , and carries a payload m_p . It moves under gravity and has relevant viscous friction at both joints. In the absence of a priori information on the dynamic parameters, design an adaptive control law yielding global asymptotic stabilization of the tracking error for a desired trajectory $q_d(t)$.



the model is :

$$\begin{pmatrix} m_1 & 0 \\ 0 & m_2 + m_p \end{pmatrix} \begin{pmatrix} \ddot{q}_1 \\ \ddot{q}_2 \end{pmatrix} + \begin{pmatrix} 0 \\ (m_2 + m_p) g_0 \end{pmatrix} = \begin{pmatrix} \tau_1 \\ \tau_2 \end{pmatrix} - \begin{pmatrix} b_1 & 0 \\ 0 & b_2 \end{pmatrix} \begin{pmatrix} \dot{q}_1 \\ \dot{q}_2 \end{pmatrix}$$

We don't know b_1 and b_2 , so the friction terms. First of all we express the model in the form $Yp = z$: $\leftarrow q_r = q_d + \begin{pmatrix} \lambda_1 & \lambda_2 \end{pmatrix} (q_d - q)$

$$\begin{cases} m_1 \ddot{q}_1 + b_1 \dot{q}_1 = \tau_1 \\ (m_2 + m_p)(\ddot{q}_2 + g_0) + b_2 \dot{q}_2 = \tau_2 \end{cases} \Rightarrow \underbrace{\begin{pmatrix} \dot{q}_{1r} & 0 & m_1 \ddot{q}_{1r} \\ 0 & \dot{q}_{2r} & (m_2 + m_p)(\ddot{q}_{2r} + g_0) \end{pmatrix}}_Y \underbrace{\begin{pmatrix} b_1 \\ b_2 \\ 1 \end{pmatrix}}_{\hat{p}} = z$$

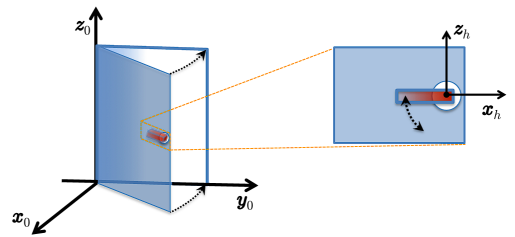
we will update b_1, b_2 , we set:

$$z = Y \hat{p} + K_p e + K_D \dot{e} \quad \text{with } K_p, K_D > 0 \quad \text{and } e = q_d - q$$

$$\text{then } \hat{p} = \Gamma Y^T (\dot{q}_r - \dot{q}) \quad \text{with } \Gamma > 0.$$

Exercise #5

A robot should close a door by firmly grasping its handle, pushing the door to the final position, and then turning the handle to lock the door. This interaction task is sketched in Fig. 3. Write down the natural and artificial constraints, defining a suitable task frame. Propose a time behavior for the hybrid references so as to cover the pushing phase and the final locking of the door. As usual in planning of hybrid force-velocity tasks, neglect any friction or environment compliance.



the Frame R_h have always (x_h, z_h) parallel to the door. The natural constraints are:

- grasp the handle, don't release it: $v_x = v_z = 0$ and $w_x = w_z = 0$
- artificial: push the door: $v_y > 0$

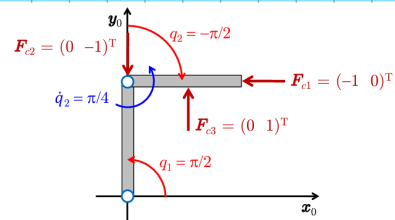
when have to lock: $w_y > 0$

$$\Rightarrow \begin{pmatrix} v \\ w \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ 1 & 0 \\ 0 & 0 \\ 0 & 0 \\ 0 & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} v_y \\ w_y \end{pmatrix} = S_t V \Rightarrow \begin{cases} \text{phase 1, pushing: } V = (\alpha, 0)^T & \alpha > 0 \\ \text{phase 2, locking: } V = (0, \beta)^T & \beta > 0 \end{cases}$$

$$W = \begin{pmatrix} f \\ n \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} f_x \\ f_y \\ n_x \\ n_z \end{pmatrix} = S_w \lambda \Rightarrow \begin{cases} \text{we don't want to stress} \\ \text{the door so we} \\ \text{set } \lambda = 0 \end{cases}$$

Exercise #6

With reference to Fig. 4, a planar 2R robot with links of unitary length (in [m]) is in the configuration $q = (\pi/2 \ -\pi/2)^T$ [rad] and with a velocity $\dot{q} = (0 \ \pi/4)^T$ [rad/s] when a collision occurs. Detail and comment the detection and isolation properties of the energy-based and momentum-based methods for the three cases of collision with the shown impact forces: i) $F_{c1} = (-1 \ 0)^T$ [N] at the end-effector; ii) $F_{c2} = (0 \ -1)^T$ [N] at the elbow; iii) $F_{c3} = (0 \ 1)^T$ [N] at the midpoint of the second link.



ii) Since the impact is on the ee, the torques produced are:

$$\tau_c = J^T F_{c1} = \begin{pmatrix} -S_1 - S_{12} & C_1 + C_{12} \\ -S_{12} & C_{12} \end{pmatrix} \begin{pmatrix} -1 \\ 0 \end{pmatrix} = \begin{pmatrix} -1 & 1 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} -1 \\ 0 \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$$

So we have: $M\ddot{q} + C\dot{q} + g = \tau + \underset{\text{actuation}}{1} \leftarrow \text{unknown}$

We build an observer using the residuals method:

$$r = K(P - \int (\tau + C\dot{q} - g + r) dt)$$

$$\dot{r} = K(\tau_c - r), \quad K = \begin{pmatrix} \kappa_1 & 0 \\ 0 & \kappa_2 \end{pmatrix} \quad \kappa_i > 0$$